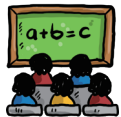


(a)



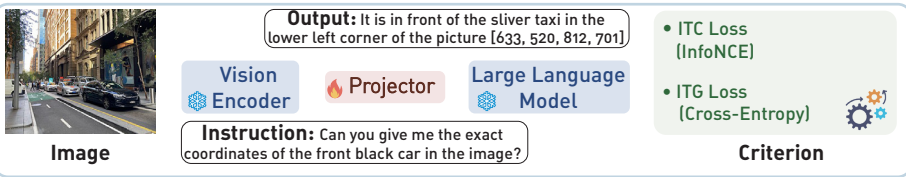
Stage I

Supervised Spatial
Concept Alignment

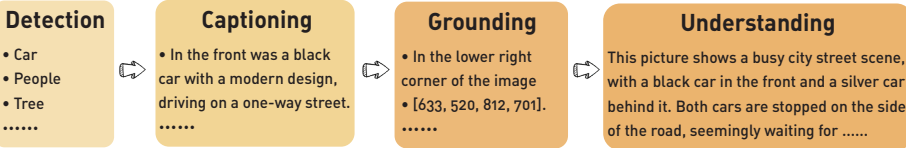
(b)



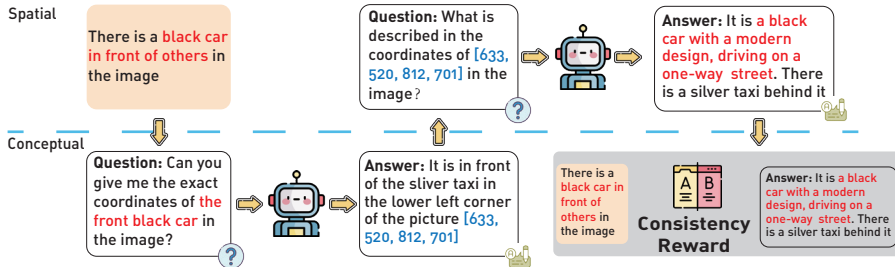
Stage II

Reinforced Explore
In the Open World

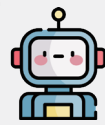
Model & Algorithm



Concept Chain of Spatial Hierarchy



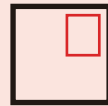
(c)



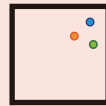
SCALAR

Vision
ModelLarge Language
Model

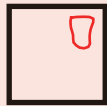
Spatial Representation



Bounding Box



Keypoints



Polygon

Conceptual Knowledge

Detection



Captioning



Grounding



Understanding

Multi-Dimensional
Joint Training / Human-AI Interactions